

Exp No: 29 PRIORITY-BASED INTERRUPT HANDLING

DATE:25/08/2026

```
kaviya@KV: ~
GNU nano 8.7.1
#include <systemc.h>

SC_MODULE(priority_interrupt) {

    sc_event low_int, high_int;

    void low_ISR() {
        while(true) {
            wait(low_int);
            cout << "Low Priority Interrupt at "
            << sc_time_stamp() << endl;
            wait(3, SC_SEC);
        }
    }

    void high_ISR() {
        while(true) {
            wait(high_int);
            cout << "High Priority Interrupt at "

            << sc_time_stamp() << endl;
            wait(1, SC_SEC);
        }
    }

    void generator() {
        wait(1, SC_SEC);
        low_int.notify();

        wait(1, SC_SEC);
        high_int.notify();
    }

    SC_CTOR(priority_interrupt) {
        SC_THREAD(low_ISR);
        SC_THREAD(high_ISR);
        SC_THREAD(generator);
    }
}
```

```

}

void high_ISR() {
while(true) {
wait(high_int);
cout << "High Priority Interrupt at "

<< sc_time_stamp() << endl;
wait(1, SC_SEC);
}
}

void generator() {
wait(1, SC_SEC);
low_int.notify();

wait(1, SC_SEC);
high_int.notify();
}

SC_CTOR(priority_interrupt) {
SC_THREAD(low_ISR);
SC_THREAD(high_ISR);
SC_THREAD(generator);
}
};

int sc_main(int argc, char* argv[]) {
priority_interrupt obj("Priority_INT");
sc_start(10, SC_SEC);
return 0;
}
|

```

```

kaviya@KV:~$ nano exp29.cpp
kaviya@KV:~$ g++ -o exp29 exp29.cpp -lsystemc
kaviya@KV:~$ ./exp29

```

```

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
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```

Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s

```